

Consider the following class:

```
class Point {
private:
    int x, y;
public:
    int distance(Point q) {
        int dx = q.x - x;
        int dy = q.y - y;
        return sqrt(dx*dx + dy*dy);
    }
};
```

If we change the above class to an interface, how shall we write the above code?
Remember an interface is a class without data members.

```
//interface.
class Point {
public:
    virtual double distance(Point* q) = 0;
    virtual int getX() = 0;
    virtual int getY() = 0;
};
```

10

// ~~Example~~ ~~usage~~ concrete use of class.

```
class Point: public Point {
private:
    int x, y;
public:
    double distance(const Point& q) const override {
        int dx = q.getX() - x;
        int dy = q.getY() - y;
        return sqrt(dx*dx + dy*dy);
    }
    int getX() const override {
        return x;
    }
    int getY() const override {
        return y;
    }
};

virtual ~Point() {}
```